HemOnc FHIR Regimen Catalog

Executable Knowledge Layer for Oncology Treatment Protocols

Beyond OMOP: what HemOnc's additional tables add — with worked examples for FOLFOX4 and a chemo + targeted kinase inhibitor combination

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Developed in cooperation with HemOnc.org · OHDSI Oncology · HL7 Europe PHOENIX · MII Onkologie Working Group

Not a medical device. Not for clinical use without independent validation.

1. Two layers of HemOnc, only one of them widely known

The HemOnc Knowledge Base (HemOncKB) is the canonical oncology vocabulary in the OMOP Common Data Model, distributed via the OHDSI Athena terminology service since 2019 (Warner JL et al., *J Biomed Inform*; HemOnc Dataverse). What most consumers see is the **OMOP core** — six tables that follow the OMOP vocabulary convention (three flat content tables plus three data-model tables that constrain them). The core is what makes HemOnc queryable from observational research stacks (OHDSI ATLAS, patient-level prediction tooling, the OMOP Common Data Model). But OMOP triples cannot represent dosing schedules, target-AUC computations, multi-component protocols, or evidence-graph relationships — and those live in the **16 additional tables** that ship alongside the OMOP core and that, until now, have not had a structured machine-readable consumer interface. This IG is that consumer interface.

Two layers, two consumers

Layer	Tables	Rows (snapshot 2026-03-12)	Typical consumer
OMOP core (content)	concept_stage, concept_relationship_stage, concept_synonym_stage	~800 k	OHDSI research sites, observational analytics, Athena
OMOP core (data model)	hemonc_classes, hemonc_rels, canonical.triples	463	Validators, schema enforcement, IG authors
Additional tables	16 tables across 5 functional groups	~310 k	This IG — clinical-ordering, CDS, evidence linkage

1.1 The OMOP core in detail

Six tables, all open and already shipped through OHDSI Athena. The content trio carries the vocabulary; the data-model trio carries the schema that constrains it.

Table	Rows	Role	What it adds
concept_stage	130 335	Content	One row per HemOnc concept (Component, Regimen, Variant, Sig, Study, Reference, Author, …). The vocabulary itself.
concept_relationship_stage	511 782	Content	Typed edges between concepts: 'Has variant', 'Was studied in', 'Maps to' (external vocabularies), 'Has cytotox chemo Rx', 'Has first author', etc. ~140 distinct relationship types.
concept_synonym_stage	159 534	Content	Alternative names per concept; supports search and string-matching against external text sources.
hemonc_classes	148	Data model	Defines every concept class with description, OMOP domain mapping, primary source table, and deprecation date.

Table	Rows	Role	What it adds
hemonc_rels	95	Data model	Defines every relationship type with heritability flag and deprecation status.
canonical.triples	220	Data model	Allowed (class■ — relationship — class■) triples — the schema constraint that turns the triple store into a typed graph (e.g., 'Regimen — Has variant — Regimen Variant').

Top OMOP concept classes by count (from `concept_stage`): Author 53 777, PMID 10 227, ReferenceDOI 10 127, Reference 9 320, Study 7 277, Regimen Variant 6 847, Brand Name 5 102, Clinical trial ID 4 328, Synthetic Regimen 3 461, Regimen 2 872, Regimen Stub 1 676, Cycle Sigs 1 317, Component 896, Regimen Class 859, Component Class 572, Endpoint 536. The 16 additional tables in the next section are the source of much of this content — `refs.csv` generates the Reference rows, `authors.csv` the Author rows, `variants.csv` the Regimen Variant rows, and so on.

2. What the additional tables carry, organized by purpose

The 16 additional tables group into five functional clusters. Together they carry the content that makes HemOnc clinically actionable — not just terminology, but executable protocols with full provenance.

Executable protocol content — *the heart of the catalogue*

Table	Rows	Adds
regimens	2 794	Regimen-family metadata: type, modality, component count, study count, highest evidence, first studied year, last published date
variants	27 393	Per-(study, regimen, variant) full encoding of the dosing protocol. Self-contained — no diffs, no inheritance
sigs	24 555	Atomic dosing unit: drug, dose (mg/m², AUC, mg/kg), route, day pattern, frequency, cycle length, infusion duration, role, relative sequencing
variant_eligibility	635	Dose-eligibility criteria: age, weight, lab thresholds

Drug knowledge — *beyond a code list*

Table	Rows	Adds
drugs	873	INN, ATC, main_class (Platinum agent, Taxane, Anti-PD-1, …), CanMED axis, investigational + multi-agent flags
rxnorm_map	901	HemOnc ↔ RxNorm crosswalk with semantic type (IN, SCD, …)
indications	3 549	Regulatory approvals by component-within-regimen and condition. Six curated regulators: FDA (1 697), EMA (636), PMDA Japan (540), MFDS South Korea (293), HC Health Canada (286), NMPA China (97). Carries date, accelerated flag, withdrawn flag, biomarker, prior-therapy, demographics, narrative reason

Disease conditions — *with eight crosswalks*

Table	Rows	Adds
conditions	318	HemOnc disease concepts with mappings to NCIt, SNOMED CT, OncoTree, ICD-9-CM, ICD-10-CM, ICD-O-3, ICD-O-3 morphology, SEER

Evidence base — *the EBM backbone*

Table	Rows	Adds
studies	7 697	Clinical trials: registry (CT.gov, EudraCT, …), trial ID, condition, biomarker, enrolment, phase, arms, design, sponsor, group
study_results	14 727	RCT endpoints with HR/OR/RR, CI bounds, p-values, comparator code, landmark — maps to FHIR EffectEvidenceSynthesis
refs	11 259	Publications: PMID, DOI, PMCID, title, journal, biblio, pub-date, ref type, errata, citation count
authors	180 345	Per-publication author records with order, role, affiliation
persons	53 545	Per-author rollup, normalized across publications

Curation transparency — *the epistemic honesty layer*

Table	Rows	Adds
inclusions	10 494	Publications HemOnc accepted with the reason (criteria-driven or SOP-exception)
exclusions	18 198	Publications HemOnc reviewed and rejected with the reason — bigger than inclusions, by design

Wiki linkage — *backlinks to human-readable content*

Table	Rows	Adds
pointers	21 935	Versioned URL backlinks to the HemOnc.org wiki Table-of-Contents (narrative explanations, dose-modification rules)

3. Worked example — FOLFOX4

FOLFOX4 is the canonical first-line systemic therapy for metastatic colorectal cancer and the prototype of bi-weekly cytotoxic combinations. It is in the MII Onkologie protocol code list as `FOLFOX4` and in HemOnc as regimen CUI 13829 with 5 variants across 28 studies. First studied 1995, last published 2024-11-15. *Phase 3 RCT* evidence level.

3.1 Regimen-level metadata

Field	Value
regimen_name	FOLFOX4
regimen_cui	13829
regimen_type	Component-based regimen
MII protocol code	`FOLFOX4` from `mii-cs-onko-systemische-therapie-protokolle` v2026.0.3
sact	TRUE (systemic anticancer therapy)
components	3 — Fluorouracil + Leucovorin + Oxaliplatin
studies	28
variants	5
highest_evidence	Phase 3 RCT
first_studied	1995
last_published	2024-11-15

3.2 Cycle structure

FOLFOX4 is a **14-day cycle** in which actual drug administration is concentrated on days 1 and 2; the patient then rests for 12 days before the next cycle begins on day 15. In adjuvant colorectal cancer it is given for **12 cycles (≈6 months)**; in the metastatic setting it continues until progression or unacceptable toxicity. The notation **“q14d”** means "repeats every 14 days".

Day-by-day calendar (one cycle)

Cycle day	What is administered	How it is given
Day 1	Oxaliplatin 85 mg/m², then Leucovorin 200 mg/m², then 5-FU 400 mg/m² bolus, then 5-FU 600 mg/m² continuous	Sequential same-day infusions: oxaliplatin (2 h) and leucovorin (2 h) given in parallel, then 5-FU bolus push, then 5-FU 600 mg/m² IV continuous over 22 h
Day 2	Leucovorin 200 mg/m², then 5-FU 400 mg/m² bolus, then 5-FU 600 mg/m² continuous	Same as day 1 but without oxaliplatin: leucovorin (2 h), 5-FU bolus, then 5-FU 600 mg/m² IV continuous over 22 h
Days 3 – 14	(rest)	No drug administration
Day 15 = Day 1 of cycle 2	Cycle repeats	Up to 12 cycles in adjuvant setting

Underlying sigs (the rows HemOnc actually carries)

Component	Dose	Route	Cycle days	Frequency / duration	Step (within day)
Oxaliplatin	85 mg/m²	IV	Day 1	Once on the day	—
Leucovorin	200 mg/m²	IV	Days 1 + 2	Once per day, over 2 h	—
Fluorouracil	400 mg/m²	IV	Days 1 + 2	Bolus push, ~1 min	1 of 2 (bolus first)
Fluorouracil	600 mg/m²	IV	Days 1 + 2	Continuous infusion, 22 h	2 of 2 (CI after bolus)

The two Fluorouracil rows are the same drug given two different ways on the same day: first a 400 mg/m² IV bolus push (step 1), immediately followed by a 600 mg/m² IV continuous infusion over 22 hours (step 2). This bolus-then-continuous-infusion pattern is characteristic of the "de Gramont" family of 5-FU regimens, of which FOLFOX4 is one. Each row corresponds to one FHIR `ActivityDefinition` resource (`kind = MedicationRequest`, referencing a `MedicationKnowledge`); the four ActivityDefinitions are orchestrated by one `PlanDefinition` for the variant. Each carries the dose as `doseQuantity { value, unit: 'mg/m2', system: UCUM }`; at `$apply` time the Library's BSA helper multiplies by the patient's BSA to produce the absolute mg dose, while the inter-row sequencing (bolus before CI) is expressed via `PlanDefinition.action.relatedAction` (`relationship = after-start`).

3.3 Regulatory indications

Regulator	Date	Condition	Stage	Prior therapy
FDA	2002-08-09	Colorectal cancer	Metastatic	after FOLFIRI
FDA	2004-01-09	Colorectal cancer	(any)	—

Each row becomes a `PlanDefinition.useContext` entry plus a `relatedArtifact[type=documentation]` pointing at the regulatory label. Biomarker-conditional approvals (KRAS-WT, BRAF V600E) also live in this table — they generate `PlanDefinition.action.condition` CQL expressions at the catalog level so downstream CDS Hooks responses can be biomarker-aware without the calling EHR encoding the rule itself.

Note on the absence of EMA rows for FOLFOX4. Oxaliplatin (Eloxatin) was EMA-approved in 1999 for metastatic colorectal cancer after 5-FU failure and in 2004 for adjuvant CRC, so the regimen is in routine use across the EU. The lack of EMA rows in HemOnc's `indications.csv` for the FOLFOX4 regimen reflects two HemOnc-internal curation choices, not a regulatory absence: (a) indications are tied to the *component-within-regimen* pair rather than to the drug alone, so an EMA approval of oxaliplatin would need explicit regulatory language naming FOLFOX4 to land in this row; (b) HemOnc's EMA curation lags its FDA curation overall (636 EMA rows vs 1 697 FDA rows table-wide). This IG simply projects what HemOnc currently encodes; closing the gap is upstream curation work, fileable as a HemOnc Dataverse issue.

3.4 Supporting studies (selected, by appearance frequency)

Study	Trial ID	Condition	Phase	First publication
MOSAIC	—	Colorectal cancer (adjuvant)	Phase 3	André T et al., NEJM 2004
NCCTG N9741	—	Metastatic colorectal cancer	Phase 3	Goldberg RM et al., JCO 2004
EORTC 40983 (EPOC)	NCT00006479	CRC liver metastases	Phase 3	Nordlinger B et al., Lancet 2008
PRIME	NCT00364013	Metastatic CRC (KRAS-WT)	Phase 3	Douillard JY et al., NEJM 2013
EACH	NCT00471965	Hepatocellular carcinoma	Phase 3	Qin S et al., JCO 2013
IDEA	—	Colon cancer (adjuvant)	Phase 3	Grothey A et al., NEJM 2018

Every study maps to a FHIR `ResearchStudy` with the registry ID as `identifier`. Every published endpoint result maps to `EffectEvidenceSynthesis` (HR, CI, p). The PlanDefinition carries `relatedArtifact[type=derived-from]` pointing at each supporting study and at the primary publication via DOI/PMID. **This is the closing-the-loop instrumentation** — every administered dose at a downstream site traces back, via the knowledge artifact, to the trial that justified it.

4. Worked example — chemo + targeted kinase inhibitor combination

mFOLFOX6 + Lenvatinib + Pembrolizumab (HemOnc cui 28874) is a contemporary five-component regimen combining a classical cytotoxic backbone with a multikinase inhibitor (Lenvatinib, ATC L01EX08, hits VEGFR-1/2/3, FGFR-1/2/3/4, RET, KIT, PDGFRα) and a checkpoint inhibitor (Pembrolizumab, anti-PD-1 mAb). Phase 3 RCT evidence, first studied 2021, five HemOnc components, three sact modalities (cytotoxic, targeted, immune). Demonstrates that the same FHIR catalog mechanics scale from a 1995-era cytotoxic protocol to a 2021-era multi-modality regimen with no structural changes to the IG.

4.1 The six sigs, side-by-side

Component	Class	Dose	Route	Day pattern	Cycle
Fluorouracil	Fluoropyrimidine	400 mg/m²	IV bolus	day 1	14 d
Fluorouracil	Fluoropyrimidine	2 400 mg/m²	IV CI	day 1	14 d
Leucovorin	Folate-reduction agent	400 mg/m²	IV	day 1	14 d
Oxaliplatin	Platinum agent	85 mg/m²	IV	day 1	14 d
Lenvatinib	Multikinase inhibitor (TKI)	8 mg	Oral	continuous	42 d
Pembrolizumab	Anti-PD-1 antibody	400 mg	IV	day 1	42 d

Two cycle lengths in one regimen (q14d chemo + q42d Lenvatinib/Pembro) is a common modern pattern that the catalog handles natively via nested `PlanDefinition.action` groups with independent `timingTiming`. The dose-target paradigm differs by component: mg/m² for cytotoxics (multiply by BSA), absolute-mg for Lenvatinib (no patient scaling) and Pembrolizumab (the FDA-approved fixed dose), with provenance preserved per-component via the `sig-target-type` CodeSystem.

4.2 Dose-modification surface for Lenvatinib

Lenvatinib has the most clinically active dose-modification rule of any drug in this regimen: starting 8 mg PO daily continuous, reduced to 4 mg every other day after grade 3 adverse events, with discontinuation thresholds. None of this lives in HemOnc's structured tables — it is in the FDA label and in the HemOnc.org wiki narrative reached via the `pointers` backlink. In our IG architecture, dose-modification rules become a separate institution-authored CQL Library that the catalog PlanDefinition references via `relatedArtifact[type=depends-on]`. Site policy plugs in at this seam without touching the catalog content. *This is why we deliberately separate protocol-knowledge from institution-policy from order-banding from administration-recording.*

5. Why this matters — closing the loop

Today, oncology protocols live as PDFs in pharmacy binders, as proprietary order sets inside individual EHR vendors (NCCN Templates™, Epic Beacon, Cerner PowerPlans), and as free-text Protokoll fields in cancer registry submissions. There is no openly-licensed, executable, versioned, machine-actionable catalogue of oncology regimens at European or US scale. HemOncKB has the content; OMOP has the vocabulary; this IG supplies the missing piece — a FHIR R4 projection of HemOnc that downstream EHR / CDS / registry / research systems can address by canonical URL.

The strategic value is not the catalog itself. The strategic value is the round-trip: an institution administering a HemOnc-derived FOLFOX4 cycle today, with structured MedicationRequests carrying provenance back to the trial that justified the dose, produces structured MedicationAdministration data tomorrow that flows back through MII Onkologie / mCODE / PHOENIX ECCM into OMOP, where the next HemOnc curation cycle sees the real-world outcome. **Knowledge** → **care delivery** → **real-world evidence** → **knowledge.** One vocabulary on both sides of the loop, instead of two.

5.1 What is in the repository today

Artifact	Status	Notes
sushi-config.yaml + ig.ini	scaffold ready	IG-Publisher consumable; declares dependency on de.medizininformatikinitiative.kerndatensatz.onkologie 2026.0.3
8 hand-authored R4 JSON resources	validated	Carboplatin + Paclitaxel Variant #83 end-to-end (2 CodeSystem, 2 MedicationKnowledge, 2 ActivityDefinition, 1 Library, 1 PlanDefinition)
CQL Library + FHIRPath dynamicValues	validated	BSA + Calvert + provenance codes; cqf-fhir 8.4.0 quirks documented in beads memory
HAPI FHIR + cqf-fhir Docker compose	running locally	Port 8096; load + $apply round-trip green; demo Patient + Observations produce Carbo 625 mg + Taxol 323.75 mg
7 IG narrative pages	first draft	index, background, cooperation, disclaimers, versioning, $apply/CDS, phase 0
beads issue tracker	13 open issues	Phase 1A scope, MII∩HemOnc match script, PHOENIX engagement, regulatory framing, IP posture, ConceptMap, etc.

5.2 What this is not

This is not a medical device. It is reference content under MDR/IVDR and 21 CFR. It is not CE-marked, FDA-cleared, or otherwise regulatorily conformed. Integration into any system intended to influence individual patient care creates an obligation on the integrator — not on the publisher — to perform appropriate regulatory qualification and conformity assessment of that downstream system. See the IG's *Disclaimers and Intended Use* page for the full statement.

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